

JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR FRONTIER MODELS IN AN OPEN-WORLD RPG

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Paper under double-blind review

ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, which places a frontier model inside Heroes of Jin Yong, a 1996 open-world role-playing game run under DOS emulation. Its ending requires the fourteen books of the Jin Yong novels and takes a first-time human player tens of hours. The game requires reading 320×200 isometric pixel art and 16-pixel Traditional Chinese, knowledge of the novels its world is built from, and a keyboard as the only input. It has no quest log, marker or score, so the model must keep its own map and goals, and it must win turn-based battles against the game AI. Progress is scored by milestones of the game itself and, once a model finishes, by completion time. Under a 60-minute budget, in which a human speedrun finishes the game, two of 12 frontier models take the compass and enter a battle, and none gains experience or holds a book. The models stall at four points of the opening: the gate of the starting house, the entrances on the world map, the instructions of the hermit and the first battle. Four-hour sessions of 9 models gain no experience either.

1 INTRODUCTION

Frontier models solve competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and several of the benchmarks that measure it are text environments, business simulations (Backlund & Petersson, 2025; Fan et al., 2026) or interactive fiction (Phan et al., 2025). A game adds a requirement they leave out: the model has to recover its own state from raw perception, because nothing in the environment reports it. The game benchmarks that keep perception each relax another part of the problem: BALROG adds a text rendering of the state (Paglieri et al., 2025), ARC-AGI-3 removes language (ARC Prize Foundation, 2026), and MineExplorer hands the model its objectives (Ju et al., 2026). None combines raw pixels, a non-English script and self-directed play, as Table 1 shows.

We present JY-CRPG-BENCH, built on Heroes of Jin Yong, a 1996 Chinese role-playing game that we run as its original DOS binary under emulation. Its ending requires recovering the fourteen books of the Jin Yong novels from an open world, a task that takes a first-time human player tens of hours, and Figure 1 shows its world. We chose the game because it tests five capabilities of a frontier model at once.

1. The model must read the screen. The frame is 320×200 pixel art in a 45° isometric projection, and every objective, prompt and statistic on it is Traditional Chinese in 16-pixel glyphs, a script that multimodal evaluation has largely neglected (Tam et al., 2025). No text is supplied to the model.
2. The model must bring knowledge the game never states. The world is assembled from the fourteen novels, so who a character is, which faction to join and which storyline leads to a book are known to a reader of the novels and written nowhere on screen.
3. The model must set its own goals. The game keeps no quest log, draws no marker and reports no score, so the model maintains its own map and plan and decides for itself what is worth doing. ARC-AGI-3 places the same goal-setting requirement at the centre of agentic intelligence (ARC Prize Foundation, 2026).



Figure 1: The environment as the model receives it, upscaled three times without interpolation.

4. The model must act through a keyboard alone, exposed through an API that it calls as a tool. Movement runs along four diagonals, no key moves straight up the screen, and every system of the game is a menu.
5. The model must win battles. Battles are turn-based on a grid against the AI of the game, with position, range and internal energy to manage, and a party at level one can lose them.

Together these form one long-horizon task, a persistent world with sparse and delayed reward over thousands of dependent steps. This is the setting of reinforcement learning, and here a frontier model plays it in context, with no training on the game. Two further properties make the game a usable benchmark. It is menu-driven, so a model that deliberates between actions is not penalised as in a real-time game (cf. Zhang et al., 2025a), and its progression state is measurable per character, so progress is scored against events of the game itself.

The next section reviews these benchmarks, Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports which milestones the models reach and how they play.

Table 1: Game benchmarks for language-model agents. Observation is what the model receives before any upscaling, a self-directed environment supplies no task list, score or objective marker during play, and the last column is the record that verifies progress.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
JY-CRPG-BENCH	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save, replay
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules

2 RELATED WORK

2.1 GAME ENVIRONMENTS AS AGENT BENCHMARKS

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). Imgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026).

The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels with a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours with a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2022), and NetHack remains the deepest environment in use for learned agents, with runs of tens of thousands of turns (Küttler et al., 2020). Cradle and Voyager drive games through a foundation model with a skill library or a code API (Tan et al., 2024; Wang et al., 2024a).

2.2 LONG-HORIZON EVALUATION AND ITS MEASUREMENT

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the setting to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), and TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025). ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation is a documented weakness even without perception: one-shot maze navigation fails by ten cells a side for every model tested (Jiang et al., 2026), and from coordinates alone the failures are attributed to looping (Einarsson, 2025).

Long horizons also complicate measurement. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025). An audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ -bench compares the final database state with an annotated goal (Yao et al., 2024).

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

Heroes of Jin Yong, published by Heluo Studio in 1996, is an open world of two tiers: many locations such as buildings, sects and caves attached to one world map of 480×480 tiles (scarsity

& contributors, 2026). Each of 320 named characters carries a record of level, experience, health, stamina, skills, items, reputation and aptitude, and every location carries an entry condition. Of the 84 locations, 7 are open at the start: the starting house, five inns and the house of the hermit. The conversation with the hermit opens all but 4 of the rest, and from there the order of play is left to the player. The game states its goal only through its story: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. Appendix E gives the Chinese original of every name from the game.

The game fixes the order of its opening and leaves the campaign open. It starts in a small house with a fenced yard, a searchable chest, one speaking character and one exit tile, and beyond the exit the path south across the world map leads to the hermit. Figure 7 in Appendix B shows the route. A cabinet beside him holds the compass, the only source of coordinates the game offers, and his conversation opens the house of the first party member, who joins after a yes-or-no prompt at which the confirm key means no. The opening ends there and needs no battle. The campaign that follows runs through battles, the experience and levels that victories bring, and the chains of items, factions and battles that lead to the books. Battles are turn-based on a grid: each turn a character moves along the four axes and then strikes, uses an item, waits or rests, and skills draw on internal energy. The first scripted battle begins with the party drained of it, and a party that goes elsewhere first meets storylines whose battles it cannot survive at level one. Some of these battles end the game when lost, leaving only the load menu, and others let the story continue. With all fourteen books and enough reputation, the ending is a boss battle at a shrine on the far side of the world map (bocelli, 2023).

The two phases test different capabilities. The opening is played by walking, reading and answering, and we give the model the route as far as the compass, so a stall there is a failure of perception or navigation. The campaign adds combat and the knowledge of the novels, since the game names its characters, factions and storylines and explains none of them.

3.2 OBSERVATION AND ACTION SPACE

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The frame combines several documented weaknesses of vision-language models. Low resolution reduces their accuracy (Niu et al., 2025), and images far from the training distribution degrade recognition even of familiar categories (Lin et al., 2026). Text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024b). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 6 shows the few the game responds to. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses up expecting to walk up the screen moves up and to the right.

The single action presses a key, or a list of up to 100 keys in order, holding each for a requested number of emulated frames. It returns once the screen has settled, including after a change of location, so the frame the model receives is the one to act on. There is no call that advances the game without input. A scored session answers four of the calls in Table 2 and refuses the three that save, list and restore emulator snapshots, so a run cannot be rewound from outside the game. The model can use the save and load menu of the game, so a model that loses a battle can reload the last save the game wrote. Appendix B states the settle rule in frames, and Appendix C gives the reasons for the list action and for the other choices of the protocol.

3.3 HARNESS AND SESSION PROTOCOL

The benchmark supplies the brief and the control surface, and the model plays through an agent harness. Every model in this paper ran inside pi¹, a minimal coding agent, installed without modification and set to its high thinking level. Its four tools read, write and edit files and run a shell, from which the model drives the control surface with HTTP calls. The brief is the whole instruction the model receives. It covers the session call, the calls that constitute play, the opening route as far as the compass, and a manual of the controls and systems of the game. It is published as `agents.md`,

¹<https://github.com/badlogic/pi-mono>

the project file pi reads at launch, in English and in Chinese, and both versions keep the on-screen Traditional Chinese terms verbatim. An action returns only its metadata unless the model asks for the frame in the same reply, so the model looks at the screen when it decides to. The benchmark adds no memory scaffold, no map, no pathfinder and no access to the state of the game. A model keeps any memory between actions in files it writes in that shell, and a scored session receives no field that reports its progress until the run ends, so a model cannot query its own score. The harness shell has network access, and a session that reads the catalogue or the timelines of other sessions does not count. The leaderboard ranks models under one fixed harness, since harness choice changes game results (Hu et al., 2026; Karten et al., 2026b) and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b).

A session is created with a declared model name and a playtime of 240 minutes by default and at most twenty-four hours, and its clock starts at the first playable moment. The server keeps the wall clock of every decision call, so played time and think time are measured for any client. Each session is a separate emulator process with a private copy of the game, supplied by the host, with its save slots reset to a new game, so no run inherits the progress of another. A session ends when its budget lapses or the game is completed. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline. Figure 2 shows the apparatus, and Appendix B lists the emulation and hosting parameters.

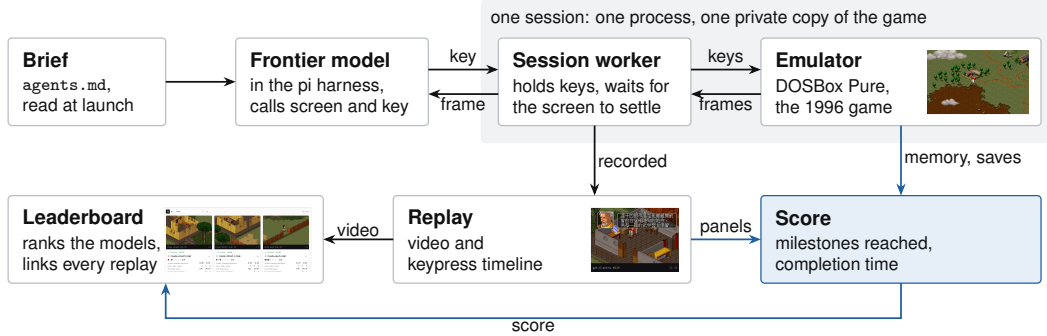


Figure 2: A frontier model in pi plays through the control API of its own session. Blue marks the measurement, which reads the memory of the emulator, the saves of the game and the panels of the replay, and which the model cannot see.

3.4 MEASURING PROGRESS

Progress is read from three records that the control surface never exposes: the memory of the emulated machine, the save the game writes, and the published replay. The game keeps its persistent state in one archive, and the shared inventory in it changes the moment an item is taken. From the live machine the benchmark reads the item count, the compass and the books after every action, and the level and experience of every character. The party roster and the world position are read from the save the game writes. The game offers its save menu on the world map alone and draws it with six rows there and four in a location, so every two minutes, once the model has sent no key for two seconds, the benchmark presses one escape and saves when the menu has six rows. A successful save is therefore the evidence that the party has crossed onto the world map, and the first one times the crossing.

Six events leave no trace in either record, because the game keeps them only on screen. The model enters a location from the world map, speaks with the hermit, opens the item screen with the compass in it, enters a battle, loses it, or answers the prompt of a party member who asks to join. Each draws fixed art at a fixed screen position, so these events are read from the published replay by matching every frame against that art. The readings do not alter the game: the benchmark serialises the machine and never loads state into it, and the only keys it presses on its own belong to the save attempt. A milestone, once observed, stays credited, so an item picked up and later used still counts. Appendix D lists the record behind each milestone, the rules of the save attempt and the panel matching, and checks the readings against one another wherever two records exist.

The milestone scale has 11 steps, following the achievements of Crafter (Hafner, 2022): six in the opening and five in the campaign. The opening milestones are the exit tile, the chest in the first room, a location entered from the world map, the conversation with the hermit, the compass in his cabinet, and the party member who joins through dialogue. The campaign milestones are a battle entered, a battle finished, experience gained, the second level, and the first of the fourteen books. The milestones are listed in the order of the opening route, with the optional chest after the exit, and a run is credited with every milestone it reaches, so the count does not depend on the path taken.

The milestones extend to the ending of the game. The fourteen books² are required before the final sequence can be played, and the archive records every book held, so the count of books continues the scale beyond the last milestone. The first reading that finds all fourteen books records the time of the last book. The ending is the conversation with the hermit after the last battle. It is read from the replay and records the completion time, which ranks the sessions that finish.

4 EVALUATION

We evaluate 12 models from 5 vendors at the 60-minute budget, which is sized to the opening, with 42 sessions on record and 2 to 6 per model. A model is credited with every milestone any of its sessions reached, and a session created under a variant of a model name, a spelling or the thinking level, is listed under the model. `claude-opus-5.5` played only four-hour sessions, and we count the first 60 minutes of each of its 2 sessions as an hour session. The service ends a session after ten minutes without an action, which stopped 11 of the 42 sessions before the hour. The baseline is a random policy that plays the same budget through the same control surface without reading the screen (Mnih et al., 2015; Hafner, 2022; ARC Prize Foundation, 2026). Its 2 sessions replay one key sequence drawn from seed 1996 and agree on every milestone, and Appendix C gives the distribution of its keys. The human references are read from published videos of the game, 4 speedruns and 3 playthroughs by experienced players, scored by the milestone readings of a replay, which Appendix D describes.

4.1 MAIN RESULTS

Of the 12 models, all 12 reach the world map, 10 pick up an item, 5 speak with the hermit, and two take the compass, enter a battle and finish it. `gpt-6-astra` also recruits a party member, reaching 8 of the 11 milestones across its 4 sessions, and `claude-opus-5.5` reaches 7 in one of its 2 sessions. No model gains experience: every one of the 37 character sheets read still shows level 1, 0 experience and 1 skill, the values at the start of the game, so every milestone from experience onward remains unreached. Figure 3 shows, for each model, the share of sessions that reached each milestone, with the human references in its two top rows. The speedruns, all of which finish the game in 60 to 126 minutes, cross onto the world map within the first minute in 32 to 107 keypresses, speak with the hermit by minute 2, enter a battle by minute 3 and hold the first book between minutes 26 and 72, and 2 of the 4 take the compass. The playthroughs, one of which reaches the ending, hold the first book between minutes 58 and 105.

4.2 QUALITATIVE ANALYSIS

The random policy reaches 1 of the 11 milestones in 60 minutes, since a random walk in a small room reaches the chest within the hour. No save succeeds during its hour, so the world map is the first milestone a random walk does not reach. The chest therefore separates models from one another but not from the baseline: 10 models searched it and 2 never did. Beyond the chest the models stall at four points of the opening.

The first is the gate of the starting house. Every model crosses onto the world map, in 32 of the 42 sessions, and 7 models cross in every session they played. The starting house is left through a yard closed by a fence with one gate, and Figure 4 draws the walk of every model through it. Several sessions were stuck in that yard. `glm-5.3-flash` pressed 1640 keys to cross in one session and reached the world map at minute 37 after long stretches of walking into the fence, while the median

²Item records 144 to 157 of the item table, named after the novels that Appendix E lists. A book counts as held in the shared inventory or carried by the player.

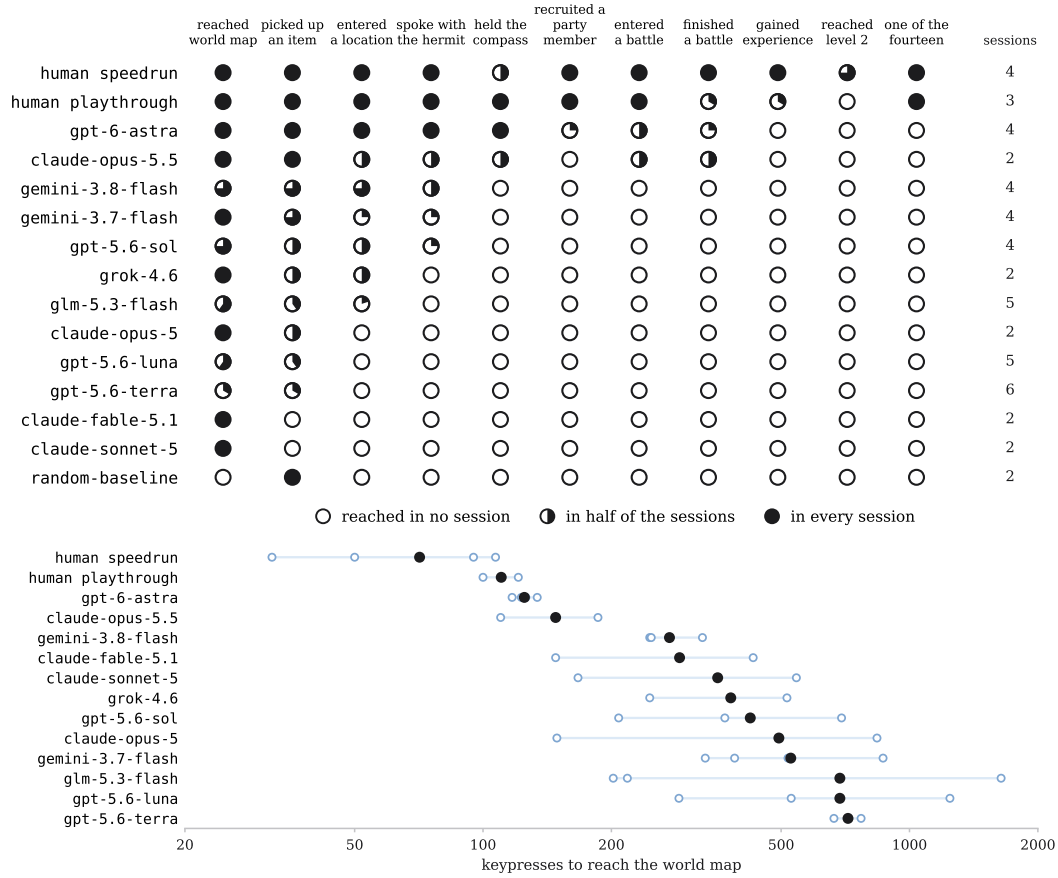


Figure 3: Milestones per model. The rows of the upper panel are ordered by the milestones reached and the rows of the lower panel by the mean keypresses to the world map, with the random baseline last. In the upper panel each marker shows the share of sessions in which the model reached the milestone, with the number of sessions at the right. The lower panel shows the keypresses to the world map across the sessions that crossed, on a log scale, with a light dot per session and a black dot for their mean. The two human rows are read from published videos, and Appendix D gives the record behind each milestone.

crossing took 308 keypresses. The count varies more within a model than between models, as the lower panel of Figure 3 shows. The crossings of `glm-5.3-flash` differ by a factor of 8 and the means of the models by a factor of 6, while `gpt-6-astra` crossed in 117 to 134 keypresses in all 4 sessions. A session of `claude-opus-5.5` crossed in 110 keypresses and 42 actions, the fewest of any model session, and reached the world map at minute 3.

The second is the entrances on the world map. After the crossing, the rest of the hour is spent there, and Figure 5 draws three walks on it. From it 7 models entered a location, for all but `gpt-6-astra` an inn or the house of the hermit, and the other 5 stayed on the world map or went back into the starting house. `claude-fable-5.1` crossed at minute 13, then spent 1261 actions and 44 minutes on the world map without entering a location, passing the gate of the hermit repeatedly, since the entrance of a walled house is one tile on one face.

The third is the dialogue of the hermit, which the model must read and act on. `claude-opus-5.5`, `gemin-3.7-flash`, `gemin-3.8-flash`, `gpt-5.6-sol` and `gpt-6-astra` reach his conversation in 9 sessions, and the conversation is long. `gemin-3.8-flash` pressed 40 times through his dialogue at 23 seconds a press, so it took 15 minutes and ended at minute 54, when the hermit had named the cabinet and 6 minutes of budget remained. `gemin-3.7-flash` reached him at minute 37 and pressed 60 times in 1 minute. Neither searched the cabinet. `gpt-6-astra` reached him in every one of its 4 sessions, between

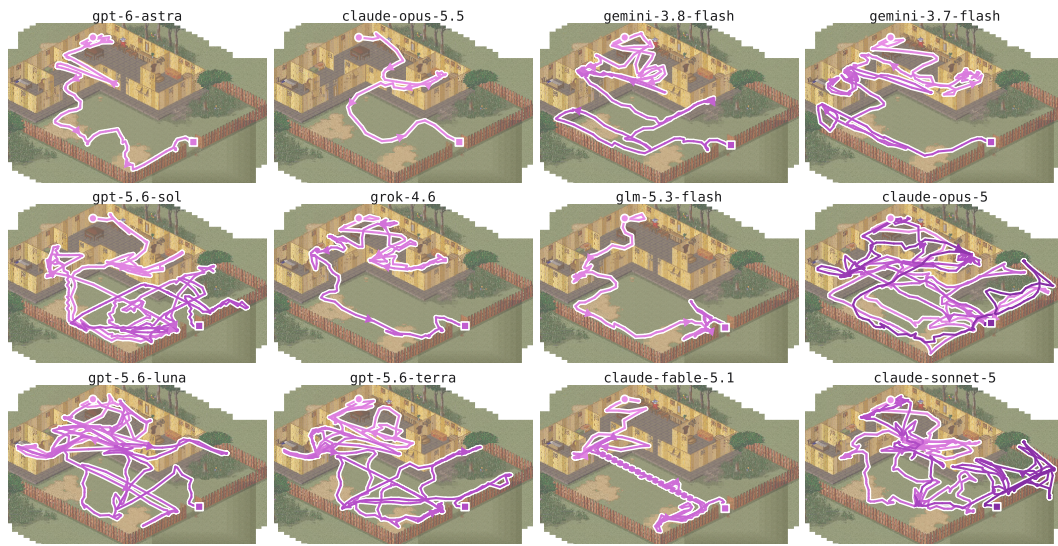


Figure 4: The walk of every model through the starting house in its session with the most milestones, from the spawn tile (circle) to the crossing (square). The colour gives the minute of play on the scale of Figure 5 and the arrowheads the direction. The panels follow the upper panel of Figure 3, and Appendix D describes how a walk is read.

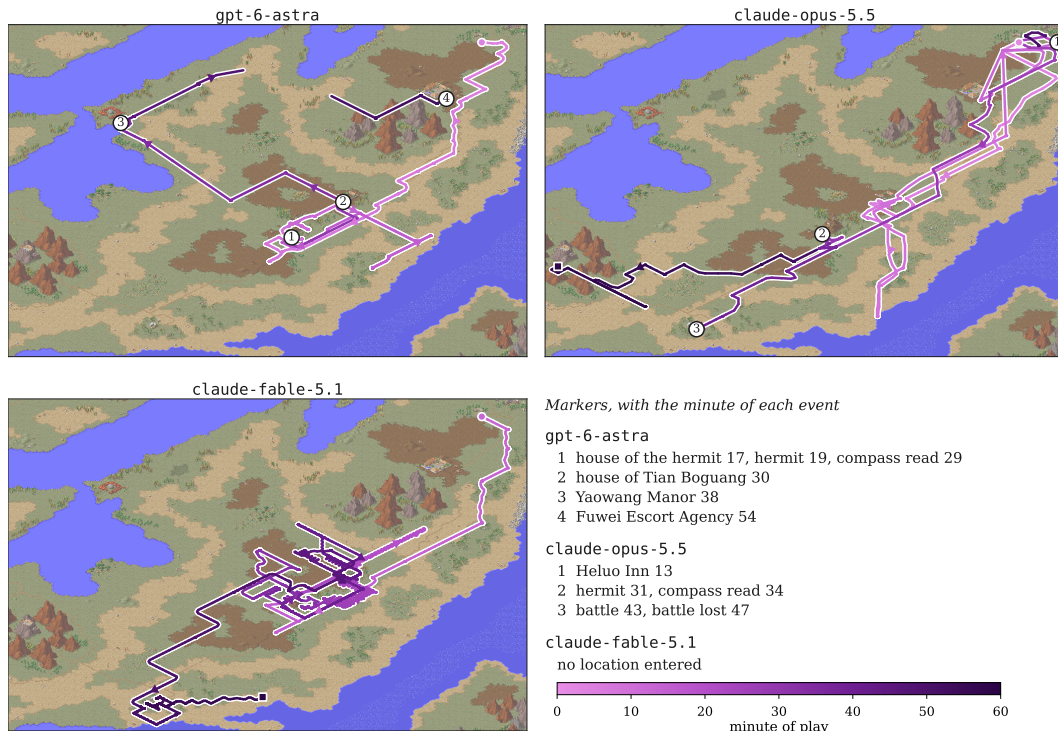


Figure 5: Three walks on the world map in the first hour, coloured by the minute of play: the session of gpt-6-astra that entered the most locations, the session of claude-opus-5.5 with the most milestones, and the session of claude-fable-5.1 that stayed longest on the world map without entering a location. A numbered marker stands at the last point on the world map before a location entered or a panel read from the replay, and the lower right lists the markers with their minutes. A gap in a path is a visit to a location or a loaded save.

minutes 11 and 19. The 4 other models that reached him did so in at most half of their sessions. The session of `gpt-6-astra` drawn in Figure 5, which entered 4 distinct locations within the hour, took 7 minutes over the conversation at 10 seconds a press and searched the cabinet. It then opened the item screen 9 times to read its coordinates from the compass, and all of its 42 arrow-key presses were made inside menus and prompts, none on the world map.

The fourth is the first battle. Two models went beyond the compass. All 4 sessions of `gpt-6-astra` read the compass between minutes 18 and 29, and diverged afterwards. One met Duan Yu at an inn at minute 52, answered yes at minute 52 when he asked to travel together, and was handed the manual of his art. One entered a battle at minute 27 and lost it at minute 28, then opened the system menu, loaded the save the benchmark had written, and by minute 34 held the compass again. One entered a battle at minute 40 and sent its last key at minute 42 with the battle still open. `claude-opus-5.5` took the compass in one of its sessions. It entered an inn at minute 13 and asked the waiter for directions, as the brief describes. It spoke with the hermit at minute 31, took the compass from his cabinet at minute 33 and read its coordinates at minute 34. At minute 43 it fought a swordsman who had heard that it sought the fourteen books, and at minute 47 it lost the battle. No model won a battle, and none reached a campaign milestone beyond it.

4.3 FOUR-HOUR SESSIONS

The 240-minute budget has 48 attempts on record, and a session counts when it played to its budget: the service ended it at the budget and its last key falls within the final 15 minutes. 10 sessions of 9 models count, and Table 5 in Appendix B gives, for each, the minute of its last key, the minute at which it reached each milestone and its milestone count. The other 38 attempts do not count, and Appendix B gives the reasons. One of them, of `gemini-3.8-flash`, played to the budget with a harness that read the catalogue and the keypress timelines of earlier sessions, and it held the compass and recruited a party member. While stuck in the opening, the model used its shell to fetch the catalogue and the timelines, ranked them by actions to the world map, and read the key sequences of the fastest, among them a session of `gpt-6-astra`. Reward hacking of this kind is common in agent traces (Shao et al., 2026), and a long horizon gives a stuck model hours to look for it. The sessions that count stay inside the opening save one: 9 of the 10 reach the world map, between minutes 2.9 and 52.4, and 5 enter a location. The session of `claude-opus-5.5` whose first hour Figure 5 draws speaks with the hermit at minute 31, reads the compass at minute 34, enters a battle at minute 43 and loses it at minute 47, and in the three hours after its first reaches no further milestone. No session gains experience or holds a book. The other 6 models with four-hour attempts, `deepseek-v4-flash`, `gemini-3.8-flash`, `gpt-5.6-terra`, `gpt-6-astra`, `grok-4.7` and `kimi-k3`, have no session that counts.

5 CONCLUSION

We presented JY-CRPG-BENCH, a long-horizon benchmark built on a 1996 open-world role-playing game, in which every model plays under the same brief and control surface and is scored against events of the game itself. No frontier model plays it reliably. At the 60-minute budget, in which a human speedrun finishes the game, every one of the 12 models reaches the world map, two, `gpt-6-astra` and `claude-opus-5.5`, take the compass and enter a battle, and none gains experience. Four-hour sessions of 9 models gain no experience either, and the one that loses a battle does so in its first hour and reaches nothing new in the next three, so more time does not carry a model further. The models fall into four traps. They circle the yard of the starting house before they find its one gate, they walk past the one-tile entrances on the world map, they read the instructions of the hermit without acting on them, and none wins a battle. The first three test perception, exploration and reading, and the fourth tests combat, which the campaign requires together with the knowledge of the novels. We plan two evaluations. Sessions of eight to twenty-four hours bring the campaign milestones within reach, and a human reference run under the same brief and budget, reported as Wei et al. (2025) recommend, calibrates the milestones against a first-time player.

AI USE STATEMENT

We used generative AI tools, coding agents built on frontier language models, throughout this work. They implemented the benchmark service, the replay scanner and the scripts that generate every number, figure and table in this paper from the session records. They recovered and reformatted session records, supported the qualitative analysis of the replays, and proposed and refined parts of the measurement design. They searched for and formatted related work, and they drafted and edited the text of this paper under our direction. We have not used generative AI tools to generate synthetic data, to formulate or prove mathematical claims, or to propose hypotheses. The models evaluated in the benchmark are the object of study. We have reviewed all AI-assisted work. Every number in the paper is regenerated from the session records at each build, every typed claim about the records is verified by a consistency script, and the frames and readings behind the qualitative analysis were checked against the replays. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The benchmark runs a commercial 1996 game that is not redistributed: every host supplies its own copy, and the frames in this paper identify the interface under study, as Appendix A explains. The work involves no human participants, and the public catalogue carries the metrics, videos and keypress timelines of model sessions and no personal data.

REPRODUCIBILITY STATEMENT

The benchmark code, the brief the models receive, the replay scanner and the scripts that generate every number, figure and table in this paper will be released with the paper. The release includes the records, videos and keypress timelines of every session in the field. Appendix B lists the emulation and hosting parameters and the dates of the sessions, Appendix C gives the reasons for the protocol choices, and Appendix D the readings behind every milestone. The game itself is not redistributable, as Appendix A states.

REFERENCES

- Anthropic. Claude’s extended thinking. Research post, section “Claude plays Pokémon”, <https://www.anthropic.com/research/visible-extended-thinking>, 2025.
- ARC Prize Foundation. ARC-AGI-3: A new challenge for frontier agentic intelligence. *arXiv preprint arXiv:2603.24621*, 2026.
- Axel Backlund and Lukas Petersson. Vending-Bench: A benchmark for long-term coherence of autonomous agents, 2025. URL <https://arxiv.org/abs/2502.15840>.
- Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The Arcade Learning Environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, 2013.
- bocelli. A complete walkthrough of the original Heroes of Jin Yong (in Chinese). <https://forum.gamer.com.tw/C.php?bsn=10859&snA=2778>, 2023.
- Mingguang Chen, Licheng Wang, and Bo Qu. The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon LLM agents, 2026. URL <https://arxiv.org/abs/2608.06663>.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025. URL <https://arxiv.org/abs/2507.06261>.
- Hafsteinn Einarsson. MazeEval: A benchmark for testing sequential decision-making in language models, 2025. URL <https://arxiv.org/abs/2507.20395>.

- Wei Fan, Xinjie Shen, Xudong Guo, Jianhong Tu, Yang Su, Yinger Zhang, Lianghao Deng, Fengyu Wang, Baohua Dong, Yangqiu Song, and Dayiheng Liu. E-Commerce Bench: Evaluating LLM agents on long-horizon autonomous business operation, 2026. URL <https://arxiv.org/abs/2608.30730>.
- Danijar Hafner. Benchmarking the spectrum of agent capabilities. In *International Conference on Learning Representations*, 2022.
- Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, Haoyang Yu, Eric P Xing, Ion Stoica, Tajana Rosing, Haojian Jin, and Hao Zhang. Igame-Bench: How good are LLMs at playing games? In *International Conference on Learning Representations*, 2026.
- Yuhan Jiang, Peng Luo, and Liqiu Meng. Lost in Aggregation: A multi-scale diagnostic benchmark for LLM spatial navigation, 2026. URL <https://arxiv.org/abs/2606.22219>.
- Tianjie Ju, Yueqing Sun, Zheng Wu, Wei Zhang, Yaqi Huo, Xi Su, Qi Gu, Xunliang Cai, Gongshen Liu, and Zhuosheng Zhang. MineExplorer: Evaluating open-world exploration of MLLM agents in Minecraft, 2026. URL <https://arxiv.org/abs/2605.30931>.
- Sayash Kapoor, Peter Kirgis, Andrew Schwartz, Stephan Rabanser, J. J. Allaire, Rishi Bommasani, Harry Coppock, Magda Dubois, Gillian K Hadfield, Andrew B. Hall, Sara Hooker, Seth Lazar, Steve Newman, Dimitris Papailiopoulos, Shoshannah Tekofsky, Helen Toner, Cozmin Ududec, and Arvind Narayanan. Open-world evaluations for measuring frontier AI capabilities, 2026. URL <https://arxiv.org/abs/2605.20520>.
- Seth Karten, Jake Grigsby, Tersoo Upaa Jr, Junik Bae, Seonghun Hong, Hyunyoung Jeong, Jaeyoon Jung, Kun Kerdthaisong, Gyungbo Kim, Hyeokgi Kim, Yujin Kim, Eunju Kwon, Dongyu Liu, Patrick Mariglia, Sangyeon Park, Benedikt Schink, Xianwei Shi, Anthony Sistilli, Joseph Twin, Arian Urdu, Matin Urdu, Qiao Wang, Ling Wu, Wenli Zhang, Kunsheng Zhou, Stephanie Milani, Kiran Vodrahalli, Amy Zhang, Fei Fang, Yuke Zhu, and Chi Jin. The PokeAgent Challenge: Competitive and long-context learning at scale, 2026a. URL <https://arxiv.org/abs/2603.15563>. NeurIPS 2025 Competition Track.
- Seth Karten, Joel Zhang, Tersoo Upaa Jr, Ruihong Feng, Wenzhe Li, Chengshuai Shi, Chi Jin, and Kiran Vodrahalli. Continual harness: Online adaptation for self-improving foundation agents, 2026b. URL <https://arxiv.org/abs/2605.09998>.
- Heinrich Küttler, Nantas Nardelli, Alexander Miller, Roberta Raileanu, Marco Selvatici, Edward Grefenstette, and Tim Rocktäschel. The NetHack learning environment. *Advances in Neural Information Processing Systems*, 33:7671–7684, 2020.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Chris Painter, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long software tasks, 2025. URL <https://arxiv.org/abs/2503.14499>. NeurIPS 2025.
- Wenhao Li, Wenwu Li, Chuyun Shen, Junjie Sheng, Zixiao Huang, Di Wu, Yun Hua, Wei Yin, Xiangfeng Wang, Hongyuan Zha, et al. TextAtari: 100k frames game playing with language agents. *arXiv preprint arXiv:2506.04098*, 2025.
- Ling Lin, Yang Bai, Heng Su, Congcong Zhu, Yaoxing Wang, Yang Zhou, Huazhu Fu, and Jingrun Chen. OODBench: Out-of-distribution benchmark for large vision-language models, 2026.
- Qing’an Liu, Juntong Feng, Yuhao Wang, Xinzhe Han, Yujie Cheng, Yue Zhu, Haiwen Diao, Yunzhi Zhuge, and Huchuan Lu. VISTA-Bench: Do vision-language models really understand visualized text as well as pure text?, 2026.
- Haotian Luo, Huaisong Zhang, Xuelin Zhang, Haoyu Wang, Zeyu Qin, Wenjie Lu, Guozheng Ma, Haiying He, Yingsha Xie, Qiyang Zhou, Zixuan Hu, Hongze Mi, Yibo Wang, Naiqiang Tan, Hong Chen, Yi R. Fung, Chun Yuan, and Li Shen. UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios, 2025. URL <https://arxiv.org/abs/2509.21766>.

- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Belle-
mare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen,
Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharmashan Kumaran, Daan Wier-
stra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning.
Nature, 518(7540):529–533, 2015.
- Junbo Niu, Yuanhong Zheng, Ziyang Miao, Hejun Dong, Chunjiang Ge, Hao Liang, Ma Lu, Bohan
Zeng, Qiahao Zheng, Conghui He, and Wentao Zhang. Native visual understanding: Resolving
resolution dilemmas in vision-language models, 2025.
- Mingyu Ouyang, Siyuan Hu, Kevin Qinghong Lin, Hwee Tou Ng, and Mike Zheng Shou. Game-
World: Towards standardized and verifiable evaluation of multimodal game agents, 2026. URL
<https://arxiv.org/abs/2604.07429>.
- Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir
Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, et al. BALROG: Bench-
marking agentic LLM and VLM reasoning on games. In *International Conference on Learning
Representations*, 2025.
- Dongmin Park, Minkyu Kim, Beongjun Choi, Junhyuck Kim, Keon Lee, Jonghyun Lee, Inkyu
Park, Byeong-Uk Lee, Jaeyoung Hwang, Jaewoo Ahn, et al. Orak: A foundational benchmark
for training and evaluating LLM agents on diverse video games. In *International Conference on
Learning Representations*, 2026.
- Long Phan, Mantas Mazeika, Andy Zou, and Dan Hendrycks. TextQuests: How good are LLMs at
text-based video games? *arXiv preprint arXiv:2507.23701*, 2025.
- Pooyan Rahmazadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision
language models are blind: Failing to translate detailed visual features into words, 2024. URL
<https://arxiv.org/abs/2407.06581>.
- scarsty and contributors. kys-cpp: a C++ reimplementation of Heroes of Jin Yong. GitHub reposi-
tory, <https://github.com/scarsty/kys-cpp>, 2026.
- Jiaqi Shao, Hanck Chen, Wei Zhang, Maxm Pan, and Bing Luo. Do agent benchmarks measure
capability? Protocol validity in the age of agentic AI, 2026. URL <https://arxiv.org/abs/2607.22368>.
- Zhi Rui Tam, Ya-Ting Pai, Yen-Wei Lee, and Yun-Nung Chen. VisTW: Benchmarking vision-
language models for Traditional Chinese in Taiwan, 2025. URL <https://arxiv.org/abs/2503.10427>.
- Weihao Tan, Wentao Zhang, Xinrun Xu, Haochong Xia, Ziluo Ding, Boyu Li, Bohan Zhou, Jun-
peng Yue, Jiechuan Jiang, Yewen Li, Ruyi An, Molei Qin, Chuqiao Zong, Longtao Zheng, Yu-
jie Wu, Xiaoqiang Chai, Yifei Bi, Tianbao Xie, Pengjie Gu, Xiyun Li, Ceyao Zhang, Long
Tian, Chaojie Wang, Xinrun Wang, Börje F. Karlsson, Bo An, Shuicheng Yan, and Zongqing
Lu. Cradle: Empowering foundation agents towards general computer control, 2024. URL
<https://arxiv.org/abs/2403.03186>.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan,
and Anima Anandkumar. Voyager: An open-ended embodied agent with large language mod-
els. *Transactions on Machine Learning Research*, 2024a. URL <https://arxiv.org/abs/2305.16291>.
- Jiayu Wang, Yifei Ming, Zhenmei Shi, Vibhav Vineet, Xin Wang, Yixuan Li, and Neel Joshi. Is
a picture worth a thousand words? Delving into spatial reasoning for vision language models,
2024b. URL <https://arxiv.org/abs/2406.14852>. NeurIPS 2024.
- Archana Warriar, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder,
Cambridge Yang, Joshua B Tenenbaum, Sebastian Vollmer, Kevin Ellis, et al. Benchmarking
world-model learning with environment-level queries. *arXiv preprint arXiv:2510.19788*, 2025.

- Kevin L. Wei, Patricia Paskov, Sunishchal Dev, Michael J. Byun, Anka Reuel, Xavier Roberts-Gaal, Rachel Calcott, Evie Coxon, and Chinmay Deshpande. Position: Human baselines in model evaluations need rigor and transparency (with recommendations & reporting checklist). In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 82265–82325, 2025. URL <https://proceedings.mlr.press/v267/wei25s.html>.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, et al. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *Advances in Neural Information Processing Systems*, 37:52040–52094, 2024.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, 2024. URL <https://arxiv.org/abs/2406.12045>.
- Alex L Zhang, Thomas L Griffiths, Karthik R Narasimhan, and Ofir Press. VideoGameBench: Can vision-language models complete popular video games? *arXiv preprint arXiv:2505.18134*, 2025a.
- Haoran Zhang, Chenhao Zhu, Sicong Guo, Hanzhe Guo, Haiming Li, and Donglin Yu. StarBench: A turn-based RPG benchmark for agentic multimodal decision-making and information seeking, 2025b. URL <https://arxiv.org/abs/2510.18483>.
- Ruizhi Zhang, Ye Huang, Yuangang Pan, Chuanfu Shen, Zhilin Liu, Ting Xie, Haijun Lei, and Lixin Duan. Mastering PokeGym: Graph-guided multimodal evolution at test time, 2026a. URL <https://arxiv.org/abs/2604.08340>.
- Yunbei Zhang, Janet Wang, Yingqiang Ge, Weijie Xu, Jihun Hamm, and Chandan K. Reddy. Stop comparing LLM agents without disclosing the harness, 2026b. URL <https://arxiv.org/abs/2605.23950>.
- Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, et al. Establishing best practices in building rigorous agentic benchmarks. *Advances in Neural Information Processing Systems*, 38, 2025.

A RELEASE AND LICENSING

The benchmark code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2 and is loaded at runtime through the libretro C API. The bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright Soft-World International and Heluo Studio. It is not redistributable or tracked in the repository, and every host supplies its own copy. The copy the paper was measured with is identified by two SHA-256 digests. One is of the program file the launcher loads. The other is of a manifest over the 616 shipped files without the three rewritable save slots. The manifest has one line per file, sorted by name, with digest and name.

```
Z.DAT 0034f836def2b287e62b0902f08cab22b61b397f24d44d6038274f1750a007ce
manifest 29ef75ab32a7905e0b274a9bba9e9ea0ff1b081e74aeacedf86350604aedc558
```

A copy with another layout yields no reading, as Appendix C explains. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The frames of the game in this paper identify the interface under study, as prior work showing frames of copyrighted games has done (Paglieri et al., 2025). If the rights holders object, they will be removed.

B INTERFACE AND SESSION DETAILS

Table 2 lists the control surface, and Figure 6 shows the keyboard the API exposes. A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 of 1996, with no sound card, so the observation is visual only. The published video runs at 8 times speed, or faster for long runs, with a strip naming

the model, the action number, the keys held and the clock, beside a timeline file with every keypress and its hold. The hour sessions ran from 10 to 12 September 2026 and the four-hour sessions, whose first hours give the hour sessions of `claude-opus-5.5`, from 16 to 22 September 2026. Each session is listed under the model that played it, with the thinking level omitted since every session ran at the high level; the name declared to the catalogue differs for 8 four-hour sessions, and the release records the mapping. Table 3 lists the effort of every model in the quantities the service records: sessions, actions, keys per action and the time between actions. The budget is wall clock, so the compute a model spends on a decision and the latency of its provider are charged against the time it has left to play, as ARC-AGI-3 and TextQuests charge it (ARC Prize Foundation, 2026; Phan et al., 2025). The sessions took between 31 and 1994 actions, and the pace did not set the order: the model that reached the most milestones is neither the fastest between actions nor the one that took the most actions in the hour. How often a model looked did not set it either. Of the 24 sessions whose record counts the screen reads, the median read the screen 1.00 times per action, and the rank correlation of the milestones reached is -0.17 with the keys per action over the 42 model sessions and -0.28 with the reads per action over those 24. `gpt-5.6-terra` sent lists of up to 100 keys in one action, 22 on average, so it looked at the screen once for every 22 keys it pressed. The catalogue also holds sessions the paper leaves out: 2 of `gemini-3.6-flash`, a model with two later versions in the field, and the 38 four-hour attempts that do not count: 5 sent no action, 3 sent one or two, 27 sent their last key before minute 225, 2 were declared under names that identify no model of the paper, and one read the timelines of earlier sessions. The release lists each attempt with its reason. Table 4 gives each milestone in actions and in minutes, the medians over the sessions that reached it, read from the replay and its keypress timeline.

Each key is held for a requested number of emulated frames, ten by default, because the game samples its keyboard once per loop. The action returns once the screen has settled: the game has 30 frames to react, and the wait then ends after 9 frames without change, when the picture cycles as in an idle animation, or after 120 frames regardless. A black frame marks a change of location, and the action waits through it and settles once more. A longer animation returns a frame in motion, and the next screen request shows the settled picture. Near the edge of a map the background stops scrolling and the sprite moves across it, so a model that infers movement from the background miscounts its tiles there, and canopies and roofs can hide the character.

Table 2: The control surface. A check marks a call the model can use in a scored session; the snapshot calls serve development.

Call	Meaning	Model can use
GET <code>/api/screen</code>	the current frame	✓
GET <code>/api/help</code>	the brief, in English or Chinese	✓
GET <code>/api/keys</code>	every accepted key name	✓
GET <code>/api/slots</code>	emulator snapshots on disk	×
POST <code>/api/key</code> <code>{"key": "kp3", "hold": 10}</code>	press one key, or several in order	✓
POST <code>/api/save</code> <code>{"name": "..."} </code>	take a snapshot	×
POST <code>/api/load</code> <code>{"name": "..."} </code>	restore a snapshot	×

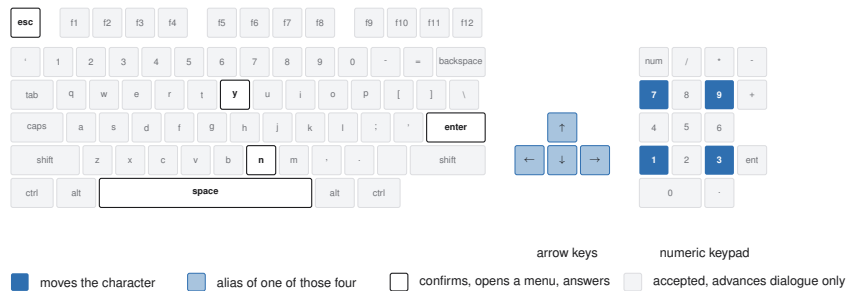


Figure 6: The keyboard the API exposes, with the nine keys the game responds to outside dialogue highlighted; any key advances a dialogue.

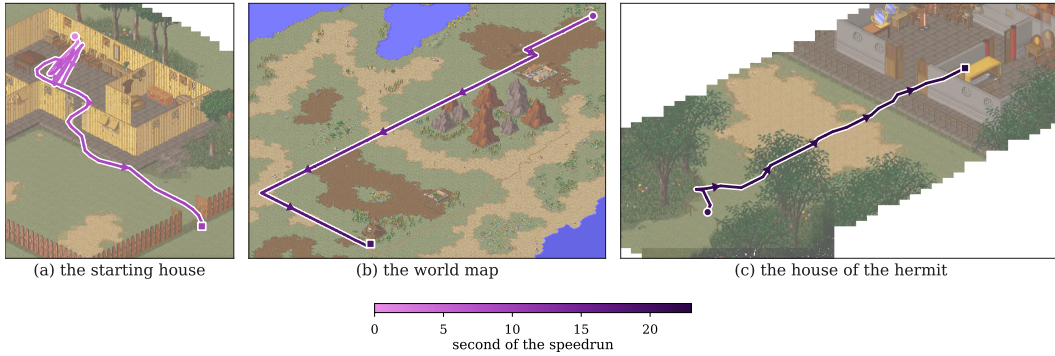


Figure 7: The opening route as far as the hermit, from a human speedrun: (a) the starting house from the spawn tile to the gate, (b) the world map from the gate to the house of the hermit, and (c) that house from its door to the hermit. The walks are read and drawn as in Figures 4 and 5, coloured by the second of the speedrun; the house of the hermit is stitched from the frames of the speedrun.

Table 3: Effort per model in the order of the upper panel of Figure 3; actions and seconds are medians over sessions.

Model	Sessions	Actions per session	Keys per action	Seconds between actions
gpt-6-astra	4	254	3.7	11
claude-opus-5.5	2	960	1.6	2
gemini-3.8-flash	4	174	3.1	18
gemini-3.7-flash	4	170	4.9	16
gpt-5.6-sol	4	332	3.1	8
grok-4.6	2	472	1.4	5
glm-5.3-flash	5	215	1.8	14
claude-opus-5	2	303	6.5	12
gpt-5.6-luna	5	155	6.9	11
gpt-5.6-terra	6	126	11.8	12
claude-fable-5.1	2	874	1.4	10
claude-sonnet-5	2	369	2.8	9
random-baseline	2	2118	1.0	1

Table 4: Each milestone in actions and minutes: the sessions that reached it and the median action count and minute at which they did.

Milestone	Sessions	Median actions	Median minute
reached the world map	32	90	17
picked up an item	22	24	3
entered a location	13	145	27
spoke with the hermit	9	135	31
held the compass	5	131	24
recruited a party member	1	269	52
entered a battle	3	164	40
finished a battle	2	460	37

C DESIGN DECISIONS

An action accepts one key or a list of up to 100, because a menu path or a repeated step is one decision. A list costs one round trip and one settle wait for the whole list. The coupling between perception and action is set by how often a model looks, since an action returns no picture unless asked, and a model that sends single keys without asking for the screen is as open-loop as one that sends twenty. The cap bounds the longest list, and every list is recorded in the public timeline.

The observation is the frame the game draws, at its own resolution, since reading sixteen-pixel glyphs and an isometric map is part of the task and the hermit milestone separates the models on

Table 5: The four-hour sessions that count, one row per session: the minute of the last key, the actions sent, the minute at which each milestone was first reached, and the milestones credited. The crossing is timed by the service and the other milestones by the replay, a dash marks a milestone not reached.

Model	Last key	Actions	Map	Item	Location	Hermit	Compass	Party	Battle	Milestones
claude-opus-5.5	229.3	11850	2.9	1.2	13.2	31.2	33.6	–	43.2	7
	234.2	5310	19.1	5.2	–	–	–	–	–	2
gpt-5.6-sol	239.5	782	19.2	210.9	90.3	–	–	–	–	3
deepseek-v4.1-flash	239.8	6026	10.3	–	21.2	–	–	–	–	2
gemini-3.7-flash	231.1	584	52.4	3.6	–	–	–	–	–	2
glm-5.3-flash	238.1	372	48.6	3.2	–	–	–	–	–	2
gpt-5.6-luna	238.9	530	9.6	–	125.8	–	–	–	–	2
grok-4.6	239.4	2857	14.5	–	148.4	–	–	–	–	2
qwen3.8-27b	239.0	203	47.6	1.6	–	–	–	–	–	2
glm-5.3	238.4	378	–	45.1	–	–	–	–	–	1

that reading. A text channel or an upscaled frame would score a different task, and a fixed OCR component would put its own errors between the game and every model. Upscaling adds no information, and a model can resample its frame with the tools of its harness. The brief lists the on-screen terms in the Traditional Chinese the game draws. The benchmark scores perception, knowledge and planning together by design, and the milestone order keeps them apart in the analysis.

The scores describe the 1996 release, run as its original binary. A reimplementaion such as `kys-cpp` (scarsty & contributors, 2026) has its own timing, text rendering and behaviour, so a track built on it would be a different benchmark. The memory reader confirms the record layout before it reads anything, so a copy with a different layout yields no reading at all.

The baseline is a seeded random policy because it measures what the milestone structure yields to chance: the small first room puts the chest within reach of a random walk, and no later milestone is. Each action draws one of the 13 keys the brief teaches, the four diagonal movement keys with 67 percent of the draws and the confirm key with 12 percent, 1.1 seconds apart on average, until the budget lapses. A scripted policy would measure the skill of its author, and a trained perception policy would measure its training data, whereas the leaderboard ranks frontier models under one harness. Cradle and Voyager are harnesses, a skill library and a code API around a model (Tan et al., 2024; Wang et al., 2024a), so a run through either would rank the harness, and such runs belong on a separate track. The service records keys and frames, so the qualitative analysis reads behaviour from the replay.

D MEASUREMENT DETAILS

The world map is read from a save written there or the world-map fingerprint; the item from the inventory count or the item message; the location from the name banner drawn on entry, the starting house excepted; the hermit from his portrait; the compass from the inventory or the item screen; the party member from the party size or a yes to the recruitment prompt; the battle from the card of the acting character; a finished battle from the defeat message or experience gained; experience and level from the character sheet; and a book from the inventory or the player. The archive holds a base block with the world position, the party roster and the shared inventory, then 320 character records of 182 bytes and an item table. The memory reader locates the records in the live machine by anchoring on one character name and confirming its two neighbours at the expected stride. A save attempt begins only while no menu is open and after two seconds without a key from the model, and recurs every two minutes and once more before the budget lapses. The session is held while the attempt runs, so the model never sees the menu, and the seconds it takes are not charged to the run. Inside a location the escape opens and closes the four-row menu, and in a dialogue it advances one line, at most once in two minutes. A fingerprint of the world map taken from the frame credits the crossing for a session with no save on record.

Every replay frame is matched against the six panels by normalised cross-correlation, and Figure 8 shows frames with them. A panel counts when it scores above 0.9 for 0.8 seconds of play: two



Figure 8: Frames from the replays behind the readings and the walks of Section 4, each with the strip naming the model, the action count and the clock: (a) the compass coordinates on the item screen; (b) the conversation with the hermit; (c) the recruitment prompt of Duan Yu in an inn; (d) a battle; (e) the hermit naming the cabinet; (f) the yard behind the fence; (g) passing the gate of the hermit; (h) talking to an innkeeper; (i) the defeat message.

consecutive frames of a replay at 8 times speed and one frame of a replay at 24 times. The game shows a panel for a second or more, so a shorter match is a screen transition. The item message varies in width with the name of the item, so it is matched along its row, and a single frame counts. A conversation advanced inside one key list shows no panel and is credited from the save, which records the locations it opened, and a recruitment prompt counts as a recruitment when the first key pressed after it appears is a yes. The walks of Figures 4, 5 and 7 place every frame of a replay or video at the offset of highest normalised cross-correlation on a panorama of the starting house, stitched from the replays, or on the world map, rendered from the data files of the game. The world map scrolls to keep the hero at one screen position, which a compass reading of a replay fixes, so the offset of a frame gives his tile.

The human rows of Figure 3 are read from 7 published videos, 4 speedruns and 3 playthroughs, listed in the release with the frame behind every reading. A capture is admitted when the opening room of a benchmark replay matches it at 0.90 or above, which locates the game frame and excludes the remakes of the game. The capture is scaled to the native frame and matched against the panels as a replay is, over a window of two pixels and at a threshold set per video in the gap of its score distribution. The clock starts where a benchmark session starts, on the first frame in which the hero stands on his spawn tile with his back to the camera, and the crossing is the first black frame after it. An action is a tile step, read from the shift of the background on the lattice of the isometric tile, or a screen change without a step, such as a line of dialogue. A won battle and a level are read from the messages that announce experience gained and a new level, a book from the item message that names it, and the ending from the closing conversation with the hermit. A recruitment is read

from the party list, from the manual the first party member hands over on joining, or from a battle in which a party member acts.

The readings agree wherever two records exist. The item message agrees with the inventory on all 36 sessions that have both and supplies the item milestone for the 8 sessions with no inventory reading. Of the 12 crossings that carry both a save and a fingerprint, the two agree on 11, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it. The crossing count read from the replay agrees with the count the service recorded on all 20 sessions that carry both. Of the 21 sessions with both an inventory reading and a replay, every one of the 3 whose inventory holds the compass shows the coordinate line on the replay, and no other session does. Of the 30 sessions with a preserved save and a replay, every one of the 3 whose save shows the locations the hermit opens also shows his portrait on the replay. No frame without a panel scored above 0.71, and every event has a frame above 0.90, so any threshold between the two gives the same readings. The first reading of every panel in the hour sessions, 20 in all, was also checked by eye against its frame, and each shows the panel.

E GLOSSARY

Table 6 gives the Chinese original of every name from the game that the paper uses, and of the fourteen books, with the English titles of the novels they are named after.

Table 6: Names from the game as the paper writes them and as the game writes them.

In the paper	In the game	Book	In the game
Heroes of Jin Yong	金庸群俠傳	The Young Flying Fox	飛狐外傳
Heluo Studio	河洛工作室	Fox Volant of the Snowy Mountain	雪山飛狐
Soft-World International	智冠科技	A Deadly Secret	連城訣
the starting house	王居	Demi-Gods and Semi-Devils	天龍八部
the hermit	南賢	The Legend of the Condor Heroes	射鵰英雄傳
the house of the hermit	南賢居	White Horse Neighs in the Western Wind	白馬嘯西風
inn	客棧	The Deer and the Cauldron	鹿鼎記
Heluo Inn	河洛客棧	The Smiling, Proud Wanderer	笑傲江湖
house of Tian Boguang	田伯光居	The Book and the Sword	書劍恩仇錄
Yaowang Manor	藥王莊	The Return of the Condor Heroes	神鵰俠侶
Fuwei Escort Agency	福威鏢局	Ode to Gallantry	俠客行
compass	羅盤	The Heaven Sword and Dragon Saber	倚天屠龍記
internal energy	內力	Sword Stained with Royal Blood	碧血劍
Duan Yu	段譽	Blade-dance of the Two Lovers	鴛鴦刀
the manual of his art	六脈神劍譜		
experience gained	獲得經驗點數		
a new level	升級了		
item message	得到		
defeat message	戰鬥失敗		